
Long Range Radar Altimeter

LR-D1 Pro

User Manual

V1.9

3/17/2024

Revision History

Version Number	Date	Notes
V1.0	June 1, 2022	Initial Release
V1.1	June 15, 2022	Added indicator information, radar installation method, calibration protocol, and alarm information.
V1.2	June 30, 2022	Revised installation instructions.
V1.3	September 1, 2022	Revised alarm protocol, startup time, power supply voltage, power rate change information.
V1.4	September 15, 2022	Added acceleration direction.
V1.5	November 15, 2022	Revised refresh rate, distance measurement accuracy above 4 meters, 60GHz output data format, alarm code description, and add CAN protocol and interface data format description.
V1.6	March 29, 2023	Revised Product Description, Product Test Tool instructions; Appendix A; Finalize Manual
V1.7	April 3, 2023	Revised Hardware interface and cabling sections.
V1.8	April 28, 2023	Updated all specifications for the release of HW V3.2. Added Firmware version section.
V1.9	March 17, 2024	Updated Firmware version information

Copyright 2023 © Ainstein AI, Inc. All rights reserved. No part of this work may be reproduced, published, or distributed in any form or by any means (electronically, mechanically, photocopying, recording, or otherwise), or stored in a database retrieval system, without the prior written permission of Ainstein AI, Inc. in each instance.

Warning!

FAILURE OR IMPROPER SELECTION OR IMPROPER USE OF THE PRODUCTS AND/OR SYSTEMS DESCRIBED HEREIN OR RELATED ITEMS CAN CAUSE DEATH, PERSONAL INJURY AND PROPERTY DAMAGE.

- This document and other information from Ainstein AI Inc, its subsidiaries and authorized distributors provide product and/or system options for further investigation by users having technical expertise.
- The user, through its own analysis and testing, is solely responsible for making the final selection of the system and components and assuring that all performance, endurance, maintenance, safety and warning requirements of the application are met. The user must analyze all aspects of the application, follow applicable industry standards, and follow the information concerning the product in the current product catalog and in any other materials provided from Ainstein or its subsidiaries or authorized distributors.
- To the extent that Ainstein or its subsidiaries or authorized distributors provide component or system options based upon data or specifications provided by the user, the user is responsible for determining that such data and specifications are suitable and sufficient for all applications and reasonably foreseeable uses of the components or systems.
- The Buyer and/or end-user will be solely responsible for final disposal of all products.

"This product is being sold as a commercial off-the-shelf radar altimeter and is not licensed or approved for any other purpose."

Table of Contents

Title Page	1
Revision History	2
Table of Contents	4
1. Product Description	6
1.1 LR-D1 Pro	6
1.2 Compliances	6
1.3 Product Features	6
2. Installation Guidelines	7
2.1. Mounting Angle	7
2.2 Mounting to an Aircraft	7
2.3. Line of Sight Clearance	7
2.4. Integration Requirements	8
3. Operational Requirements	9
3.1 Obstruction to the LR-D1 Field of View	9
3.2 Excessive Pitch/Roll	9
3.3 Terrain	9
3.4 Power Source	9
3.5 Orientation of LR-D1 Pro	9
3.6 Minimum and Maximum Operating Altitude	10
4. Technical Specifications:	11
5. Data Protocol For LR-D1 Pro:	13
6. Mechanical Drawing:	16
7. Radiated Emissions	16
8. Hardware Interface	17
9. Cabling	19
10. Software Instructions	20
10.1 Connecting the LR-D1	20
10.2 Firmware Update Instructions	22
10.3 Changing Interface Type	24
11. Firmware Version History	27
12. Contact Us	27
Appendix A - Malfunction Alerts	28

This Page Intentionally Left Blank

1. Product Description

1.1 LR-D1 Pro

The LR-D1 Pro Radar Altimeter uses the principles of radio detection and ranging to determine the altitude of an aircraft. The LR-D1 features both a 24 GHz and 60 GHz radar system, providing precise altitude measurements in takeoff and landing as well as in-flight up to 500 m in altitude. A microwave signal is transmitted out of the sensor, reflected off the target, and received by the sensor. Distance to the terrain and vertical velocity are derived by the difference in time from when the signal is sent from the sensor to when the signal is received by the sensor and by the Doppler effect.

1.2 Compliances

FCC and CE certifications pending.

1.3 Product Features

The LR-D1 Pro contains several unique features that provide advantages to the user. The first of which is the dual-band antenna design. The LR-D1 Pro utilizes both a 24 GHz and 60 GHz frequency system simultaneously. This allows for drastically finer precision below 5 m in altitude from the 60 GHz system while maintaining a range of up to 500 m from the 24 GHz system. A carefully designed algorithm switches between the two frequency systems to give the user the most precise above-ground-level (AGL) measurement possible. The redundancy of data at low altitudes makes for extra safety at critical landing or sea-skimming maneuvers of flight, which occur at 5 m altitude and below.

The LR-D1 Pro also features solid-state electronics and cutting-edge patch antennas that allow the radar to have a smaller form factor and mass compared to traditional altimeters.

The LR-D1 outputs instantaneous measurements at a high update rate of 15 ms (66 Hz), enabling the user to operate at airspeeds of 200 m/s and beyond.

The LR-D1 Pro features a rugged mechanical design that maintains the radar from being affected by adverse weather, allowing for use on aircraft that operate in rain, snow, sea spray, over water, fog, or dust. Due to the FMCW radar employed, the altimeter can output accurate measurements in these conditions, whereas laser altimeters, sonar, and cameras cannot, making the LR-D1 Pro an ideal choice for aircraft that operate over bodies of water or in extreme conditions.

The radome design of the LR-D1 Pro is specially tested to be radiative to radar signal transmission of both 24 GHz and 60 GHz frequencies, so customer customization of the enclosure is not recommended.

The LR-D1 Pro can connect and transmit data using CAN, RS-232, RS-422, RS-485, and TTL protocols, making integration into the user's flight controller as simple and straightforward as possible.

2. Installation Guidelines

2.1. Mounting Angle

When mounting the device, the front face of the LR-D1 Pro must be **perpendicular** to the ground below the aircraft. There should not be any angle of inclination in any direction while the device is fastened to the aircraft.

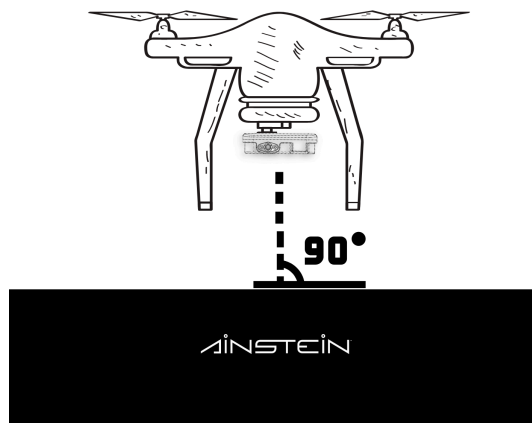


Figure 1: Mounting Angle of the LR-D1 Pro

2.2 Mounting to an Aircraft

The device should be secured to the aircraft, where it is not free to move in any direction, or exposed to vibration in excess of 15 g's.

2.3. Line of Sight Clearance

Keep the face of the radar unobstructed and do not cover it with any additional materials. Any coatings, coverings, and modifications to the radome can degrade the performance of the radar device.

Additionally, keep any unexpected objects out of the radar's FoV (Field of View). Obstructions to the LR-D1 Pro's field of view will cause a decrease in the performance of the radar. It is highly recommended that the LR-D1 Pro be mounted on the underside of the aircraft far away from the landing gear, other aircraft structures, or other equipment.

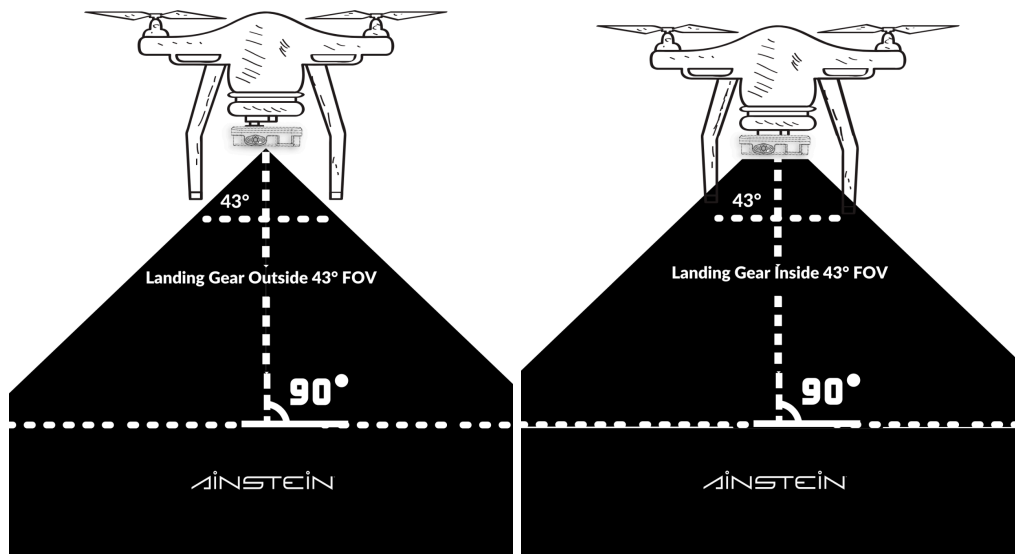


Figure 2: Line of Sight Clearance of the LR-D1 Pro

2.4. Integration Requirements

The LR-D1 Pro outputs altitude measurements and signal-to-noise ratio (SNR) measurements when operational. When integrating the LR-D1 Pro radar altimeter, it is recommended to use SNR measurements in conjunction with malfunction alerts (see Appendix 1) to properly filter out erroneous altitude values.

Caution! Altitude measurements associated with an SNR value of 13 dB or lower are considered erroneous.

A fusion algorithm should be used to estimate vehicle position, velocity, and angular orientation based on rate gyroscopes, accelerometer, compass, GPS, airspeed, and barometric pressure measurements in addition to the recorded LR-D1 Pro measurements. Sensor redundancy is strongly advised for use of the LR-D1 Pro in mission-critical application

3. Operational Requirements

The LR-D1 Pro will perform optimally if the operational requirements below are fully satisfied. Failure to meet the operational requirements may cause a decrease in performance, accuracy, or reliability of the LR-D1 Pro altimeter and is not advised.

3.1 Obstruction to the LR-D1 Field of View

Objects or aircraft structures that are located within the LR-D1 Pro's conical field of view which obstruct the radar's view of the ground may cause multipath reflections or other degradative phenomena to occur. The LR-D1 Pro should be mounted a safe distance away from the landing gear and any other components of the aircraft below it.

3.2 Excessive Pitch/Roll

Pitch and roll angles that exceed 21.5° and 15° respectively may also cause the performance of the LR-D1 Pro to worsen. This sensitivity increases with altitude.

3.3 Terrain

Terrain with poor reflectivity may cause the performance of the LR-D1 Pro to worsen. Flying at the limit of the LR-D1 Pro's range over dry, loose soil, such as tilled farmland, or sand is not recommended.

3.4 Power Source

Any power source used to operate the LR-D1 Pro that does not provide the minimum power and voltage can worsen the performance of the device or cause its operation to stop altogether.

3.5 Orientation of LR-D1 Pro

The LR-D1 Pro must be mounted upon the aircraft in such a manner as to be fully horizontal, its radome directly facing the ground. Any angle of inclination may degrade performance. The LR-D1 Pro will output an alert code "0800" when the angle of inclination of the radar is tilted beyond 20° during flight. If this alert is output while the aircraft is stationary on the ground, please adjust the radar orientation.

3.6 Minimum and Maximum Operating Altitude

Operating the LR-D1 Pro at altitudes below 0.3 meters and above 500 meters will result in a degradation of performance and potentially erroneous measurements. Also, any reading at an altitude of greater than 655.35m would be considered as an error or overflowed reading (see Appendix 1) .

4. Technical Specifications: Table 1: Specifications of the LR-D1 Pro

Frequency Band	24~24.25 GHz 60~64 GHz
Bandwidth	250 MHz 4 GHz
Power Consumption	< 14 W
Operating Voltage	10 ~ 32 V
Altitude Range	0.3m~500m ⁽¹⁾
Altitude Precision	$\pm 0.364\text{m}$ (>5m) ⁽²⁾ $\pm 0.075\text{m}$ (<5m) ⁽²⁾
Altitude Step Size	$\pm 0.075\text{m}$ (>5m) $\pm 0.015\text{m}$ (<5m)
Update Rate	15 ms (66 Hz)
Detection Angle Range	Azimuth 43°, Elevation 30° ⁽³⁾
Maximum Velocity	$\pm 30\text{ m/s}$ in elevation
Temp. Range	-40°C ~ +60°C
Dimensions	134mm*134mm*37.3mm (mounting bracket is NOT included)
Weight	685 g
IP Rating	Built to the requirements of IP67 ⁽⁴⁾
Vibration Rating	IEC 60068-2-6.1995 Sinusoidal vibration 15g XYZ three axis
Shock Rating	IEC 68-2-27:1987 half sine shock 100g XYZ three axis
ESD Rating	IEC 61000-4-2:2008 8K/15K contact/air Class B criterion
Start Time	2 s
Maximum Ambient Pressure Altitudes	10,000 m

Note:

1. Radar data may vary over different terrains when the radar is out of its detection range. It is recommended to mount the LR-D1 Pro at a minimum height of 0.3m.

We do not recommend using LR-D1 Pro at altitudes lower than 0.3m or greater than 500m. When the actual altitude is greater than 500m over terrain that provides specular reflection, such as concrete, the radar can operate up to 600m. The LR-D1 Pro will report **SNR = 0 and Altitude = 0m** to indicate if there is no target detected; Therefore, when the SNR = 0, the detection can be considered invalid for that data frame.

2. Range detection might be limited by the terrain composition below, the pitch and roll of aircraft, and operational requirements. The range precision indicates the lab tested/calibration results in the standard case.

LR-D1 Pro altitude data output step size is smaller than the radar's precision because of the post-processing of the raw detection data.

3. Based on mm-wave radar specs, a large angle of pitch and roll can cause errors in detection. Larger angles of pitch and roll will result in greater errors in LR-D1 Pro output.
4. The IP rating only includes the radar itself. This rating does not cover cabling interfaces or other accessories.

5. Data Protocol For LR-D1 Pro:

UART Data Protocol:

- I/O Standard: RS-232 (Default) and RS-422, RS-485, TTL and CAN (by configuration)
- Baud Rate: 115200 b/s
- Data length: 8 bits, plus one start bit and one stop bit, and no parity bit

CAN Data Protocol:

The output data frame of the CAN protocol is a standard frame format with the following characteristics:

- Identifier (ID) length: 11 bits (Standard CAN ID frame) , following CAN V2.0A standard protocol
- CAN ID: 0x055
- Baud rate: 1 Mbps
- Data segment length (DLC) of each CAN packet: 8 bits
- LR-D1 Pro 3.0 data packet length: 32 bytes, consisting of 8 bits from 4 CAN frames. See the figure below for the meaning of each bit of data of CAN data frames.

Table 2: Data Packet Definition

From	LR-D1 Pro	To	Receiver
Byte	Data	Note	
data1	0xEB	Packet Header MSB (Most Significant Bits)	
data2	0x90	Packet Header LSB (Least Significant Bits)	
data3	deviceID	Device ID Byte (Default: 0x00)	
data4	0x1C	Data packet length	
data5 data6	0x0000: Normal Others: Malfunction (1)	Malfunction Alert	
data7	high1_h (2)	24 GHz Altitude MSB	
data8	high1_l (2)	24 GHz Altitude LSB	
data9	snr1	24 GHz SNR	
data10	speed1_h (3)	24 GHz Velocity MSB	
data11	speed1_l (3)	24 GHz Velocity LSB	

data12	high2_h	60 GHz Altitude MSB (4)
data13	high2_l	60 GHz Altitude LSB
data14	snr2	60 GHz SNR
data15	speed2_h	60 GHz Velocity MSB
data16	speed2_l	60 GHz Velocity LSB
data17	high3_h	Integrated Altitude MSB
data18	high3_l	Integrated Altitude LSB
data19	snr3	Integrated SNR
data20	speed3_h	Integrated Velocity MSB
data21	speed3_l	Integrated Velocity LSB
data22	Reserved	0xFF
data23	Reserved	0xFF
data24	Reserved	0xFF
data25	Reserved	0xFF
data26	Reserved	0xFF
data27	Reserved	0xFF
data28	Reserved	0xFF
data29	Reserved	0xFF
data30	Reserved	0xFF
data31	Reserved	0xFF
data32	Checksum	Checksum: (data4+data5+...+data30+data31) bitwise-AND with 0xFF

Note:

1. Please see Appendix 1 for details about malfunction information.
2. Altitude Data Parse for 24 GHz: Altitude = (high1_h * 256) + high1_l; unit: **0.01 m (cm)**;
Type: **Unsigned**
3. Altitude Data Parse for 60 GHz: Altitude = ((high2_h * 256) + high2_l) , unit: 0.01 m (cm);
Type: **Unsigned**
4. Velocity Data Parse for 24 GHz: Velocity = (speed1_h * 256) + speed1_l; unit: **0.01m/s**;
Type: **Signed**
5. Velocity Data Parse for 60 GHz: Velocity = (speed1_h * 256) + speed1_l; unit: **0.01m/s**;
Type: **Signed**
6. Altitude Data Parse for Integrated: Altitude = (high3_h * 256) + high3_l, unit: **0.01 m**

(cm); ; Type: **Unsigned**

7. Velocity Data Parse for Integrated: Velocity = (speed3_h * 256) + speed3_l, unit: **0.01 m (cm)**

A 16-bit integer is used to indicate the LR-D1's altitude reading in centimeters, therefore the maximum valid altitude would be 65535 cm (655.35 m). If the actual altitude is greater than 655.35 m, an altitude reading overflow error would occur.

8. The figure below shows the CAN port output data of the device analyzed by the CAN Bus Tools tool. A single frame output result of the LR-D1 Pro is highlighted in the box. The single LR-D1 Pro frame is divided into 4 CAN data packets, with each CAN packet containing 8 bytes of data, giving a total of 32 bytes. The byte order is consistent with the product manual.

CAN-ID	Rx/Tx	Type	Length	Data
055h	Rx	Data	8	00 FF FF FF FF FF FF 1E
055h	Rx	Data	8	EB 90 00 1C 08 C0 01 B4
055h	Rx	Data	8	40 00 00 00 D8 47 00 00
055h	Rx	Data	8	00 D8 47 00 00 0D 01 00
055h	Rx	Data	8	00 FF FF FF FF FF FF 1F
055h	Rx	Data	8	EB 90 00 1C 08 C0 01 B4
055h	Rx	Data	8	42 00 00 00 D8 47 00 00
055h	Rx	Data	8	00 D8 47 00 00 0D 01 00
055h	Rx	Data	8	00 FF FF FF FF FF FF 21
055h	Rx	Data	8	EB 90 00 1C 08 C0 01 B4
055h	Rx	Data	8	40 00 00 00 D8 47 00 00
055h	Rx	Data	8	00 D8 47 00 00 0D 01 00
055h	Rx	Data	8	00 FF FF FF FF FF FF 1F

Figure 3: An LR-D1 Pro Data Packet

6. Mechanical Drawing:

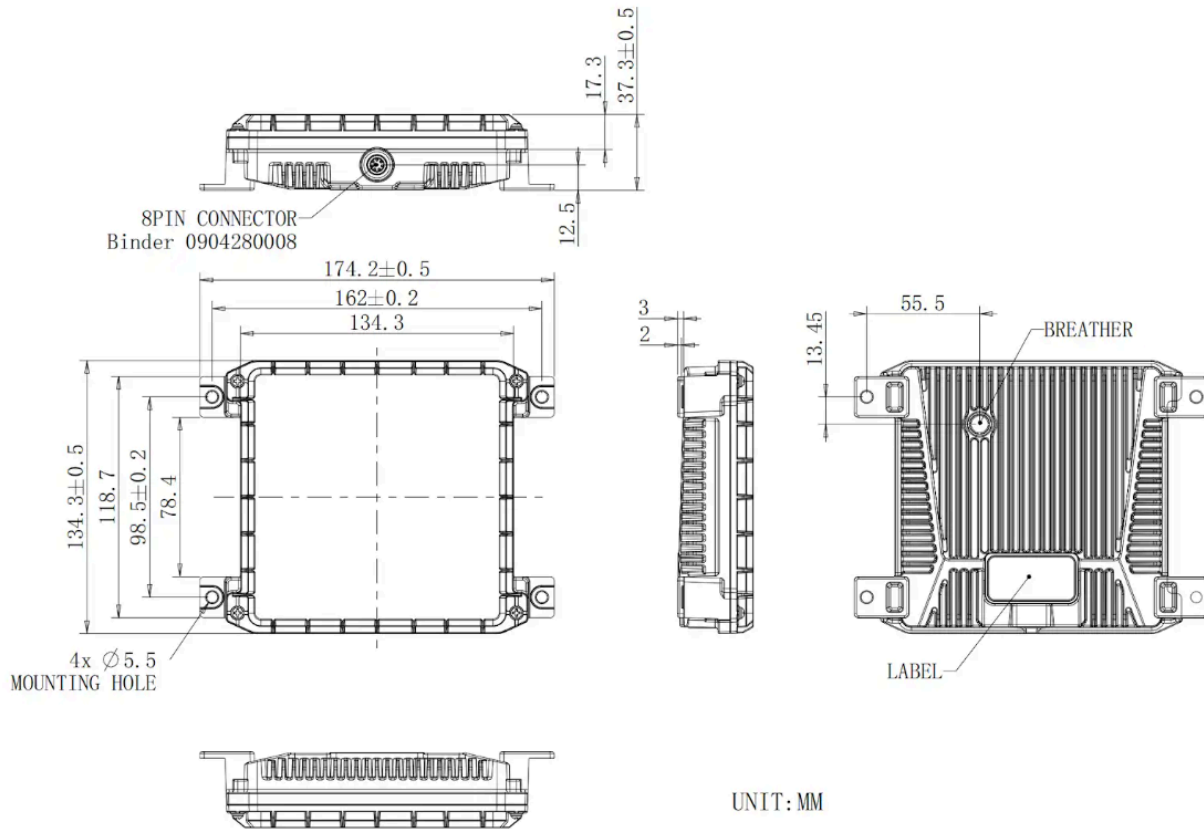


Figure 4: Dimensions of LR-D1 Pro (Units: mm)

7. Radiated Emissions

Table 3: Maximum Transmit Power

Maximum Transmit Power (EIRP)	32 dBm
-------------------------------	--------

8. Hardware Interface

There are 4 supported interfaces in the LR-D1 Pro. All 4 interfaces share the same pin definition as indicated in the table below.

Table 4: Pin Out Definition

Pin	Wire Color	Pin Name	Function	Note
1	Red	VCC	Input Voltage	10 - 32 V Power < 14W
2	Red	VCC	Input Voltage	10 - 32 V Power < 14W
3	Blue	RS-422 T+ / RS-232_T	RS422: TX+ RS232: TX CAN_H TTL: TX RS485: TX	Default Setting: RS232
4	Brown	RS-422 T- / RS-232 R	RS422: TX- RS232: RX CAN_L TTL: RX RS485: RX	Default Setting: RS232
5	White	RS-422 R-	RS422: RX-	Leave unwired for RS232
6	Green	RS-422 R+	RS422: RX+	Leave unwired for RS232
7	Black	GND	Ground	
8	Black	GND	Ground	

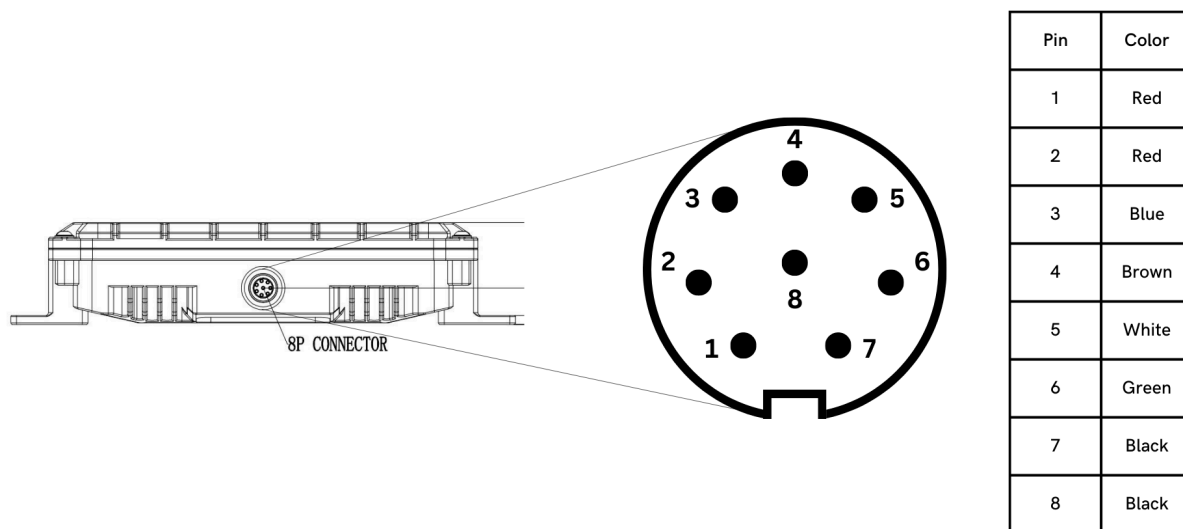


Figure 5: LR-D1 Pro Hardware Pinout Diagram

Note: Power input pins 1 and 2 share continuity and therefore a diode must be added to isolate PWR-IN-A and PWR-IN-B from each other, if isolating PWR-IN-A and PWR-IN-B is required.

9. Cabling

The default mating connector used for the LR-D1 Pro is the BINDER 99-0425-10-08 circular connector, pictured below in Figure 6.



Figure 6: 6-Wire Cable with BINDER 99-0425-10-08 Connector

The BINDER 99-0425-75-08 90° connector can be provided for an additional cost at the time of purchase. See Figure 7.



Figure 7: 6-Wire Cable with BINDER 99-0425-75-08 90° Connector

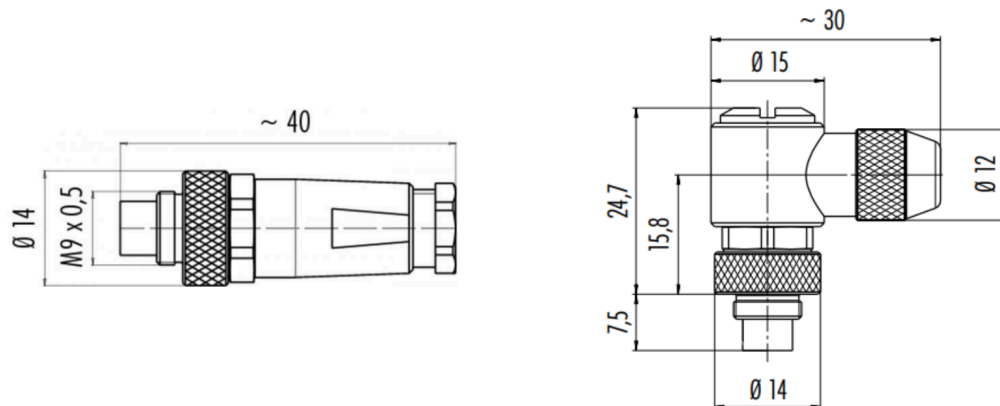


Figure 8: BINDER 99-0425-10-08 & BINDER 99-0425-75-08 Connector Dimensions.

10. Software Instructions

The settings of the LR-D1 Pro can be modified using the LR-D1 Pro Software Tool. The Software Tool can be downloaded on the “Downloads” tab of the LR-D1 Pro product page on Ainstein’s website. Instructions for connecting the LR-D1, updating the LR-D1 firmware, and changing the interface standard of the serial port are detailed below.

10.1 Connecting the LR-D1

1. After downloading the LR-D1 Pro Software Tool, open its containing folder and launch the executable, “LR-D1-PRO_V97.237.1.0”.

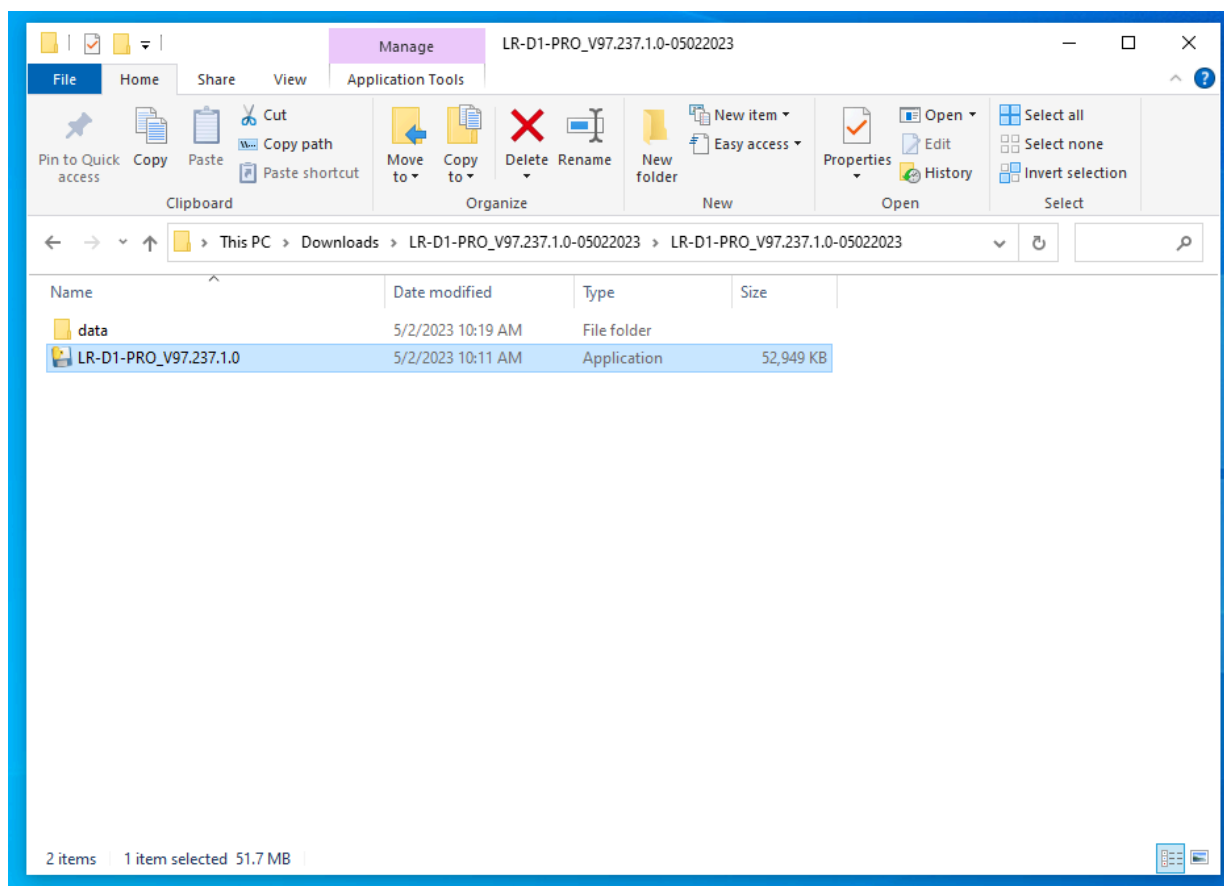


Figure 5: LR-D1 Pro Software Tool Executable

2. Once launched, navigate to the upper-left corner of the interface. In the “PORT” dropdown, select the serial port the radar is using.
3. In the “BAUD” dropdown, select a baud rate of 115200 Kbps.
4. In the “RVER” dropdown, select the version of your device. Note: If using the LR-D1 Pro, select “Gen3”, if using LR-D1, select “Gen2”.

- After ensuring the connection between the radar device and your computer over the selected serial port, click “Connect”.

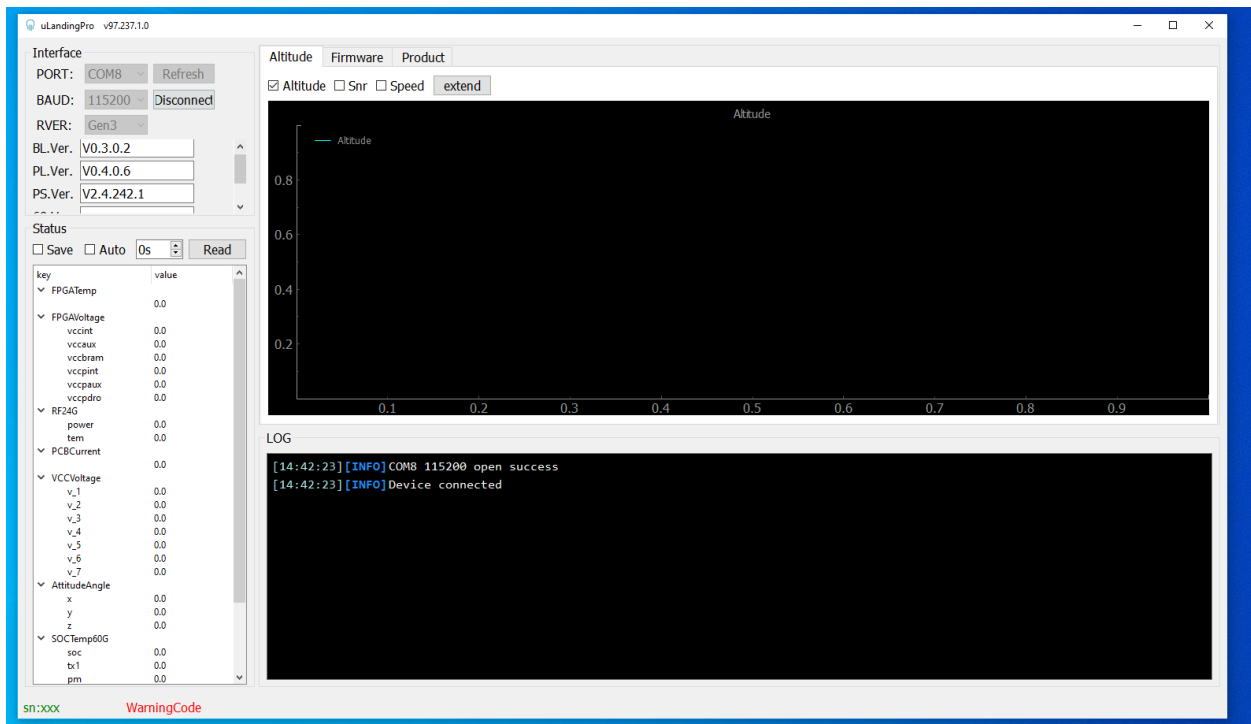


Figure 6: LR-D1 Pro Software Tool - Successful Connection

The “BL.Ver.” text field displays the bootloader version of the radar. The “PS.Ver.” text field displays the firmware version of the radar. The “Status” field on the left side of the window shows information about the radar’s temperature, voltage, current, and attitude angle. Select the “Read” button to display the current “Status” information.

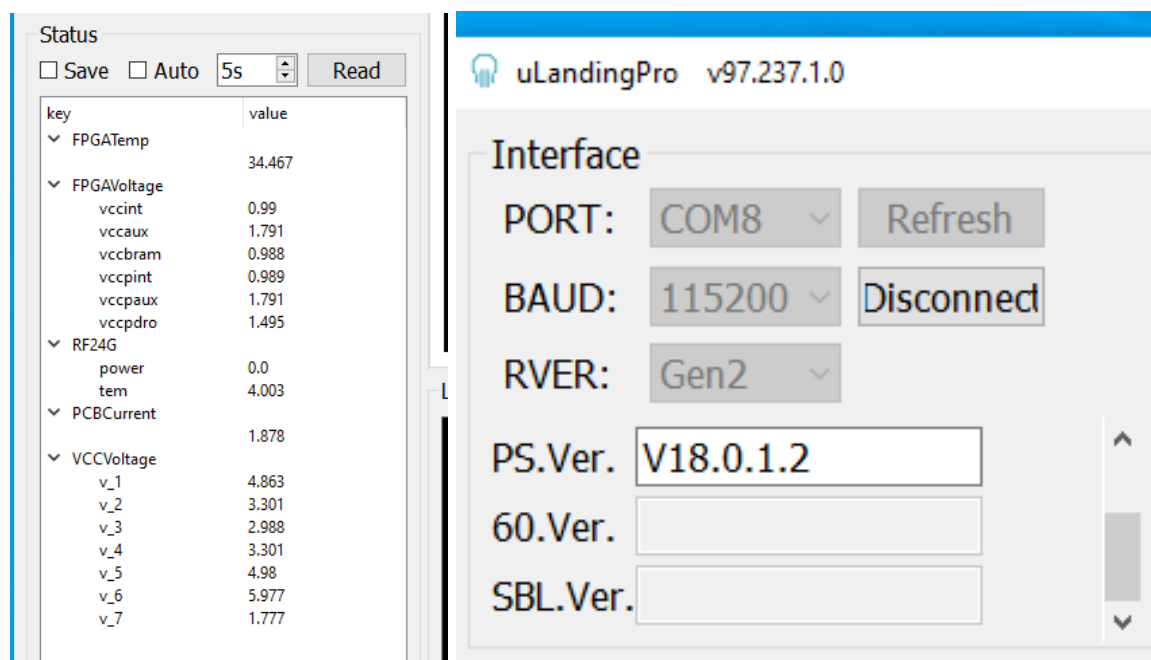


Figure 7: LR-D1 Pro Software Tool - Interface & Status Information

10.2 Firmware Update Instructions

To update the firmware version of your LR-D1 Pro device to the latest release, follow the instructions below.

1. Connect the LR-D1 Pro to the LR-D1 Pro Software Tool as described in Section 9.1.
2. Navigate to the "Firmware" tab of the main window of the tool.
3. In the "TYPE" dropdown box, select which component of the radar you would like to update. For the 24 GHz radar, select "24G", for the 60 GHz radar component, select "60G".

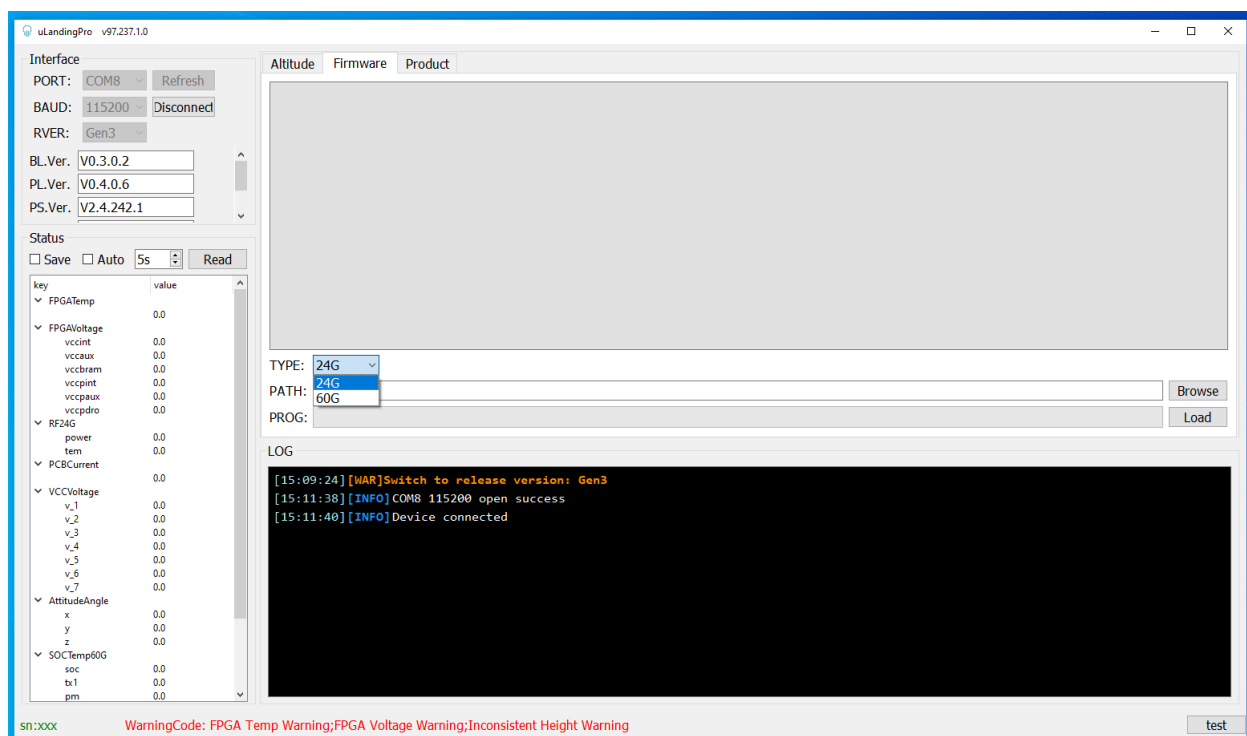


Figure 8: LR-D1 Pro Software Tool - Firmware Window

- Once selected, click the “Browse” button to select your firmware binary flash file in your PC’s directory.

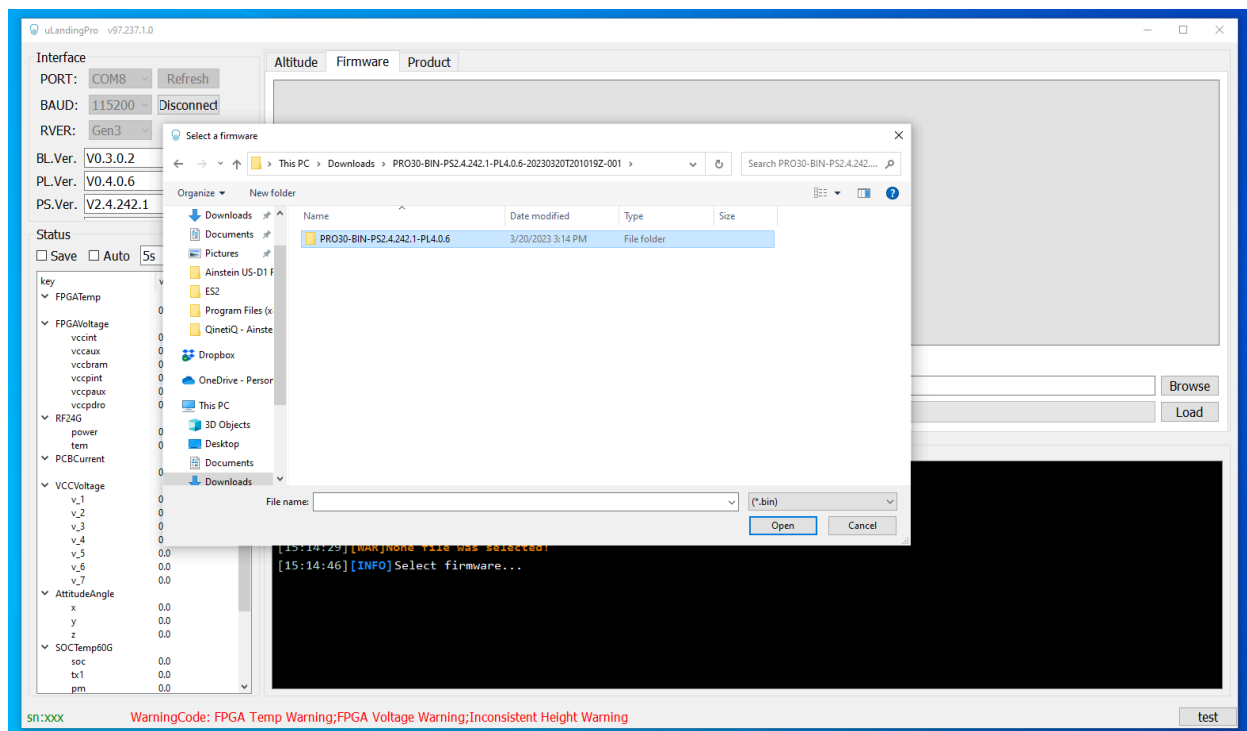


Figure 9: LR-D1 Pro Software Tool - Firmware Selection

5. For updating the 24 GHz component, select the “fpga.bin” file and click, “Open”. For the 60 GHz, select the “vave-firmware-v97.XXX.1.0.bin” file and click, “Open”. The “Firmware” window will populate with the file’s details.
6. Click “Load” to begin the firmware installation. This process may take up to 8 minutes.
7. Once completed, the LOG window will display, “Firmware update successful”. Power cycle the radar and reconnect to verify that the radar is displaying the latest firmware version.

10.3 Changing Interface Type

To change the interface type of your LR-D1 Pro device from RS-232 to RS-422, CAN, or other options, follow the instructions below.

8. Connect the LR-D1 Pro to the LR-D1 Pro Software Tool as described in Section 9.1.
9. Navigate to the “Product” tab of the main window of the tool.
10. Under the “Device” section, click the “Device Interface” dropdown. You will see a list of options.

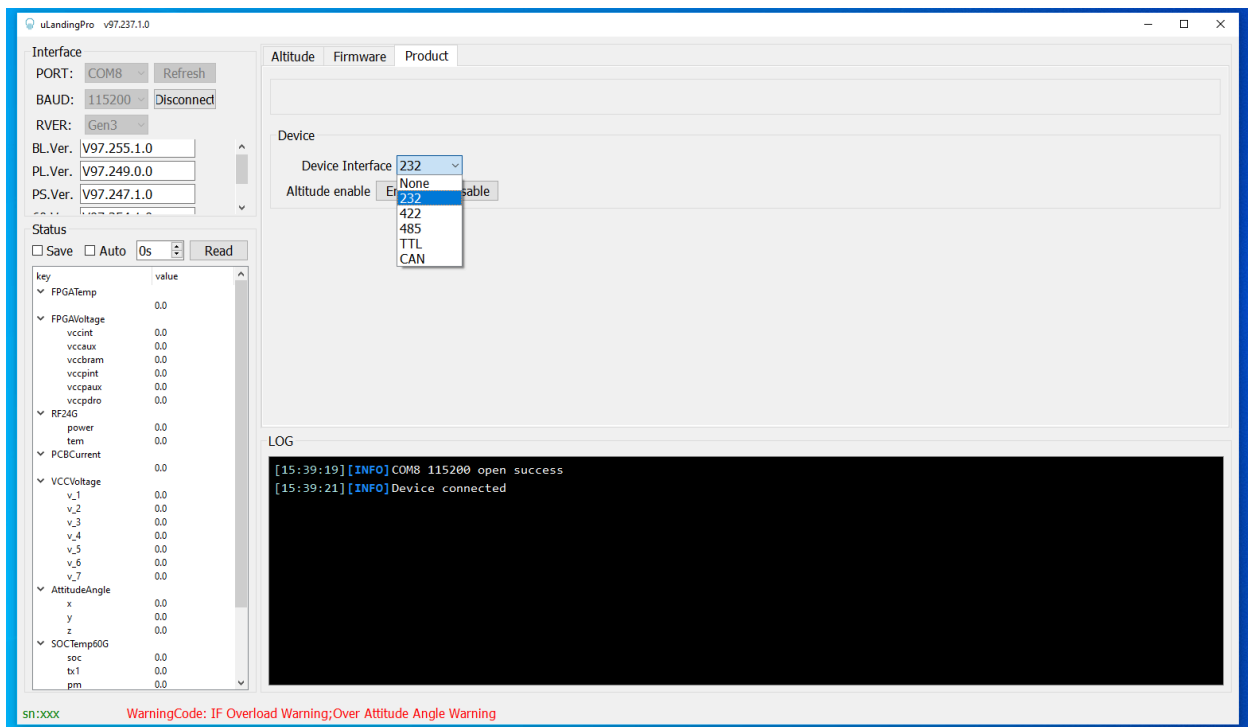


Figure 10: LR-D1 Pro Software Tool - Interface Selection

11. Select your desired interface. For RS-232, select "232". For RS-422, select "RS-422". For RS-485, select "485". For CAN protocol, select "CAN". For UART TTL, select "TTL".
12. You will be prompted by a pop-up window to modify the port to your selected protocol. If correct, select "Yes". This process will take several seconds.
13. Once completed, the LOG window will display that the communication interface setup is complete.

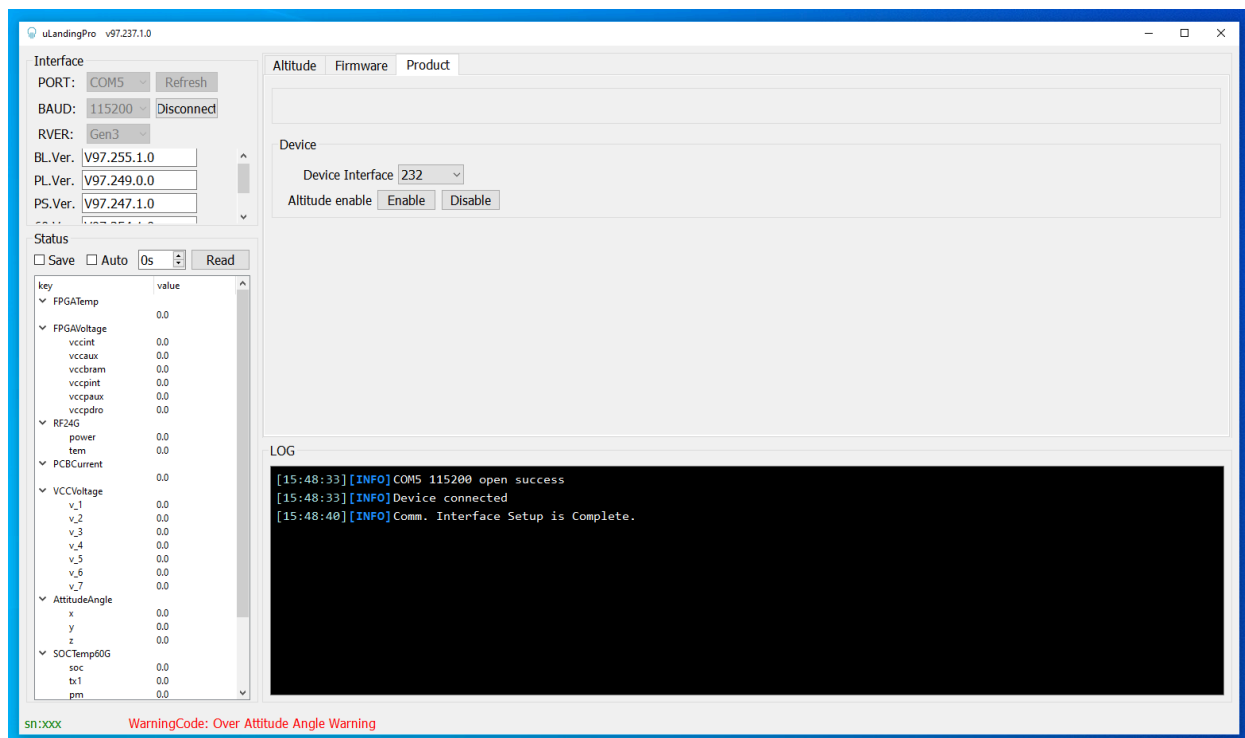


Figure 11: LR-D1 Pro Software Tool - Interface Setup Completion

11. Firmware Version History

Release Date	24 GHz Bootloader Version	PL Version	PS Version	60 GHz SBL Version	60 GHz Firmware Version
4.1.2023	V97.255.1.0	V97.249.0.0	V97.249.0.0	V97.255.0.0	V97.252.1.0
12.11.2023	V97.253.2.0	V98.255.0.3	V98.252.2.2	-	V97.251.1.0

12. Contact Us

ADDRESS:

1421 Research Park Drive
Lawrence, KS 66049

EMAIL:

hi@ainstein.ai

PHONE:

785-856-0460

Appendix A - Malfunction Alerts

The alarm code is designed as 2 bytes– a total of 16 bits– each bit representing an alarm mode. If all bits are 0, the radar is operating normally. The following 16 types of alarms exist. Data5 and data6 in the 32 bytes of data output by the device are alarm data, where data5 is the MSB and data6 is the LSB. The received two-byte alarm data needs to be converted into 8-bit hexadecimal data respectively, then compared with the hexadecimal alarm code in the table below to determine the alarm definition.

Table 5: Malfunction Alert Information

Malfunction Alert Code	Malfunction Alert Info	Possible Causes	Suggestion
0x0001	FPGA Temperature Warning	Error in the device or ambient temperature is out of LR-D1 Pro's operational temperature range.	Confirm whether the ambient temperature is beyond the LR-D1 Pro's limit, or contact Ainstein to confirm whether maintenance is needed.
0x0002	FPGA Voltage Warning	The device has an error.	Stop use and contact Ainstein for repair.
0x0004	60G-SOC Temperature Warning	Error in the device or ambient temperature is out of LR-D1 Pro's operational temperature range.	Confirm whether the ambient temperature is beyond the LR-D1 Pro's limit, or contact Ainstein to confirm whether maintenance is needed.
0x0008	60G-SOC Voltage Warning	The device has an error.	Stop use and contact Ainstein for repair.
0x0010	24G RF Temperature Warning	Error in the device or ambient temperature is out of LR-D1 Pro's operational temperature range.	Confirm whether the ambient temperature is beyond the LR-D1 Pro's limit, or contact Ainstein to confirm whether maintenance is needed.
0x0020	24G RF Transmit Power Warning	The device has an error.	Stop use and contact Ainstein for repair.
0x0040	Whole Board Overcurrent Warning	The device has an error.	Stop use and contact Ainstein for repair.
0x0080	IF Signal Saturation	May be caused by	Confirm whether

	Warning	obstacles obstructing the surface of the radar or the equipment being very close to the ground.	obstacles are blocking the surface of the radar. When very close to the ground this alert may be reported even though the device is fully functional.
0x0100	24G Software Failure Warning	The device has an error.	Stop use and contact Ainstein for repair.
0x0200	60G Software Failure Warning	The device has an error.	Stop use and contact Ainstein for repair.
0x0400	Altitude Reading Overflow Warning	Altitude reading is greater than the maximum integer of 16-bit (655.35 meters).	Occurs when LR-D1 detects the altitude is greater than 655.35 meters.
0x0800	Excessive Attitude Angle Warning	The radar tilt angle exceeds 20°.	Adjust the installation angle of the radar and ensure that the radar is perpendicular to the ground.
0x1000	60G Frame Error Warning	The device has an error.	Stop use and contact Ainstein for repair.
0x2000	Invalid Altitude Warning	Indicates that the radar has not detected a valid peak value.	Ensure the LR-D1 Pro is being used within operational requirements.
0x4000	24G & 60G Altitude Inconsistency Warning	The altitudes of the 24G and 60G antennas are inconsistent, which may be caused by RF interference.	Ensure the LR-D1 Pro is being used within operational requirements.
0x8000	Whole Board Voltage Warning	The device has an error.	Stop use and contact Ainstein for repair.